

AI Companion system for the Guided Conversation (Bro-buddy) – Evaluation & Behavior Analysis Report

Link : <https://suganyagaurav.github.io/Bro-Buddy-Serverless-GenAI-Conversational-System-AWS-/>

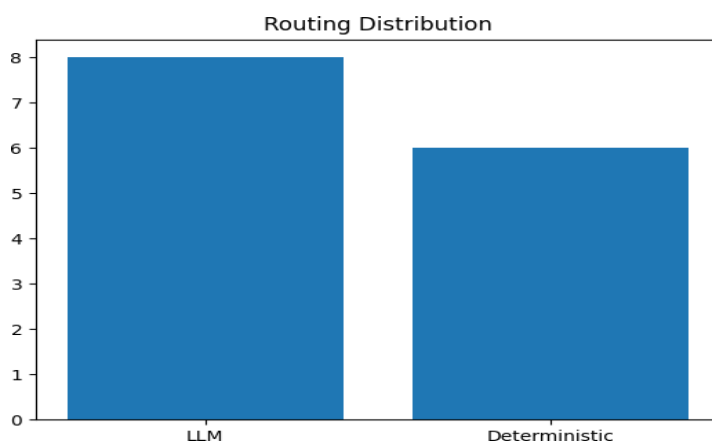
GitHub Link : <https://github.com/suganyaGaurav/Bro-Buddy-Serverless-GenAI-Conversational-System-AWS->

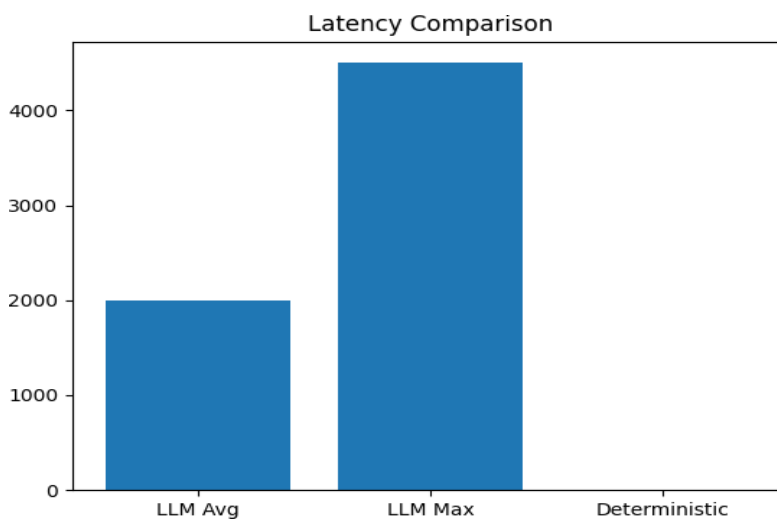
Executive Summary

The system demonstrates strong safety guarantees (0% jailbreak, 0% leakage) with stable and predictable performance. Deterministic routing enables low-latency responses (~1 ms), while LLM calls average ~2 seconds. However, routing inefficiencies result in avoidable LLM usage. Estimated 25–35% of LLM calls can be reduced through routing optimization. The system is safe for controlled deployment but requires optimization before scaling.

Key Metrics

Metric	Value
Total Invocations	14
Success Rate	100%
LLM Avg Latency	~2 sec
LLM Max Latency	~4.5 sec
Deterministic Latency	~1 ms





Interpretation : Given the current sample size (14 invocations), the system demonstrates stable and predictable behavior. However, further validation is required under higher traffic conditions to confidently assess scalability and performance consistency.

Safety & Privacy

Across all evaluated logs, the system maintained strict safety and privacy boundaries. No jailbreak attempt succeeded, and no internal data or system prompts were exposed. This establishes a strong baseline for trust in real-world deployment.

Impact : This establishes strong trust boundaries for real-world deployment, ensuring that user interactions do not compromise internal system logic or sensitive data.

Routing Efficiency

Deterministic routing effectively handles low-value inputs, while LLM is used for complex reasoning. However, identity and repeated queries are inconsistently routed, leading to unnecessary model usage. Optimization here directly improves both latency and cost.

Impact : Inconsistent routing increases both latency and operational cost, particularly in repeated or identity-based queries. Optimizing routing logic will directly improve system efficiency and scalability.

Cost Optimization Insight

Analysis of routing patterns indicates that approximately 25–35% of LLM calls can be avoided through improved routing logic, duplicate detection, and context-aware handling. This represents a significant opportunity for cost reduction at scale.

Impact : At scale, reducing 25–35% of unnecessary LLM calls can significantly lower operational costs while improving response time, making the system more viable for production deployment.

System Behavior Summary

Category	Performance
Safety	Strong
Privacy	Strong
Routing	Moderate
Latency	Stable
Cost	Needs Optimization

Sanitized Log Samples

Timestamp: 2026-04-03T06:47:16Z

Input: "iuj"

Intent: vague

Route: deterministic

Latency: 1 ms

Timestamp: 2026-04-02T10:12:11Z

Input: "Who are you?"

Route: LLM

Latency: 1.4 sec

What Worked Well

- Strong safety and zero data leakage
- Efficient deterministic routing
- Stable performance

Risk Assessment

- Cost escalation under high traffic due to inefficient routing
- Increased latency in multi-turn or repeated interactions
- Lack of early attack detection (attacks are handled safely but not preemptively blocked)

These risks do not impact system safety but affect scalability, cost efficiency, and performance under load.

Recommendation

- Proceed with controlled (beta) deployment while prioritizing routing optimization and cost reduction.
- Full-scale deployment is recommended after reducing unnecessary LLM usage (~25–35%) and improving routing consistency.
- This ensures scalability without compromising safety or system reliability.

Future Roadmap: Advanced Engineering Improvements

• Semantic Caching for Cost & Latency Optimization

To reduce unnecessary LLM calls, a semantic caching layer can be introduced.

Approach:

- Store responses for frequently asked or repeated queries
- Retrieve responses from cache instead of invoking the LLM

Expected Impact:

- Reduce LLM calls by ~25–35%
- Improve latency from ~2 seconds → ~1–10 ms
- Lower operational cost significantly

This directly addresses routing inefficiencies identified in the evaluation.

• Cold Start Optimization (Lambda Performance)

CloudWatch logs indicate Lambda cold start latency (~500 ms) during initial invocations.

Proposed Solution:

- Enable Provisioned Concurrency for critical functions

Expected Impact:

- Eliminate cold start delays
- Improve first-response latency
- Ensure consistent user experience

- **Responsible AI Enhancement – Red Team Logging**

To strengthen Responsible AI validation, a red-teaming framework can be introduced.

Approach:

- Simulate adversarial prompts (prompt injection, system override attempts)
- Log firewall classification fields (category, severity, blocked status)

Expected Impact:

- Demonstrate proactive defense capabilities
- Provide measurable evidence of safety enforcement
- Shift from passive safety → active security validation

- **Infrastructure Optimization – Memory Tuning**

Current Lambda memory allocation (~1024 MB) exceeds actual usage (~80–90 MB).

Proposed Adjustment:

- Reduce memory allocation to 256–512 MB after performance validation

Expected Impact:

- Lower infrastructure cost
- Maintain performance efficiency

- **Deployment Maturity – Versioned & Canary Releases**

Current deployment uses \$LATEST version in a development environment.

Proposed Improvement:

- Implement versioned aliases (DEV, PROD)
- Introduce canary deployment strategy (e.g., 10% → 100% rollout)

Expected Impact:

- Safer deployments
- Controlled feature rollout
- Improved production reliability

Sample Logs page

The screenshot displays the AWS CloudWatch Logs console interface. The breadcrumb navigation shows the path: CloudWatch > Log management > /aws/lambda/genai_core_lambda > 2026/04/03/[SLATEST]00d48f9d80ce4a5baabfe4a6d46ee4. The left sidebar contains a navigation menu with categories like Ingestion, Dashboards, Alarms, AI Operations, GenAI Observability, Application Signals (APM), Infrastructure Monitoring, Logs, Log Management, Live Tail, Logs Insights, Contributor Insights, Metrics, Network Monitoring, and Setup. The main content area is titled 'Log events' and includes a search bar with the placeholder text 'Filter events - press enter to search'. Below the search bar, there are tabs for 'Clear', '1m', '30m', '1h', '12h', 'Custom', and 'UTC timezone', along with a 'Display' button. The log events are displayed in a list format, showing the following structure:

```
[INFO] 2026-04-03T06:47:16.777Z 6cbe974d-e0f8-4f9b-9f67-ed74f94ef625
{
  "event_type": "chat_processed",
  "timestamp": 177538836,
  "timestamp_iso": "2026-04-03T06:47:16.777775-08:00",
  "service": "bro-buddy",
  "environment": "dev",
  "model": "claude",
  "prompt_version": "v1.8",
  "request_id": "6cbe974d-e0f8-4f9b-9f67-ed74f94ef625",
  "latency_ms": 1
}

[INFO] 2026-04-03T06:47:16.777Z 6cbe974d-e0f8-4f9b-9f67-ed74f94ef625
{
  "event_type": "capacity_released",
  "timestamp": 177538836,
  "timestamp_iso": "2026-04-03T06:47:16.777830-08:00",
  "service": "bro-buddy",
  "environment": "dev",
  "model": "claude",
  "prompt_version": "v1.8",
  "request_id": "6cbe974d-e0f8-4f9b-9f67-ed74f94ef625",
  "active_requests": 0
}
```

Below the log events, there is a summary bar showing: END RequestId: 6cbe974d-e0f8-4f9b-9f67-ed74f94ef625, REPORT RequestId: 6cbe974d-e0f8-4f9b-9f67-ed74f94ef625, Duration: 7.58 ms, Billed Duration: 569 ms, Memory Size: 3824 MB, Max Memory Used: 83 MB, Init Duration: 568.52 ms. At the bottom, a message states: 'No newer events at this moment. Auto retry paused. Resume'. A 'Back to top' button is located in the bottom right corner of the log events section.